

# Summarizing Data

View the 'big picture'

Before doing any analysis, always explore your data.

*What does the data look like?*

*Is the data appropriate for my chosen analysis?*

*How can I use it to answer my research question?*

Use both **visual** and **numeric** summaries.



The Palmer penguins dataset records physical measurements of penguins taken at Palmer Research Station.

Each row is a different penguin individual.

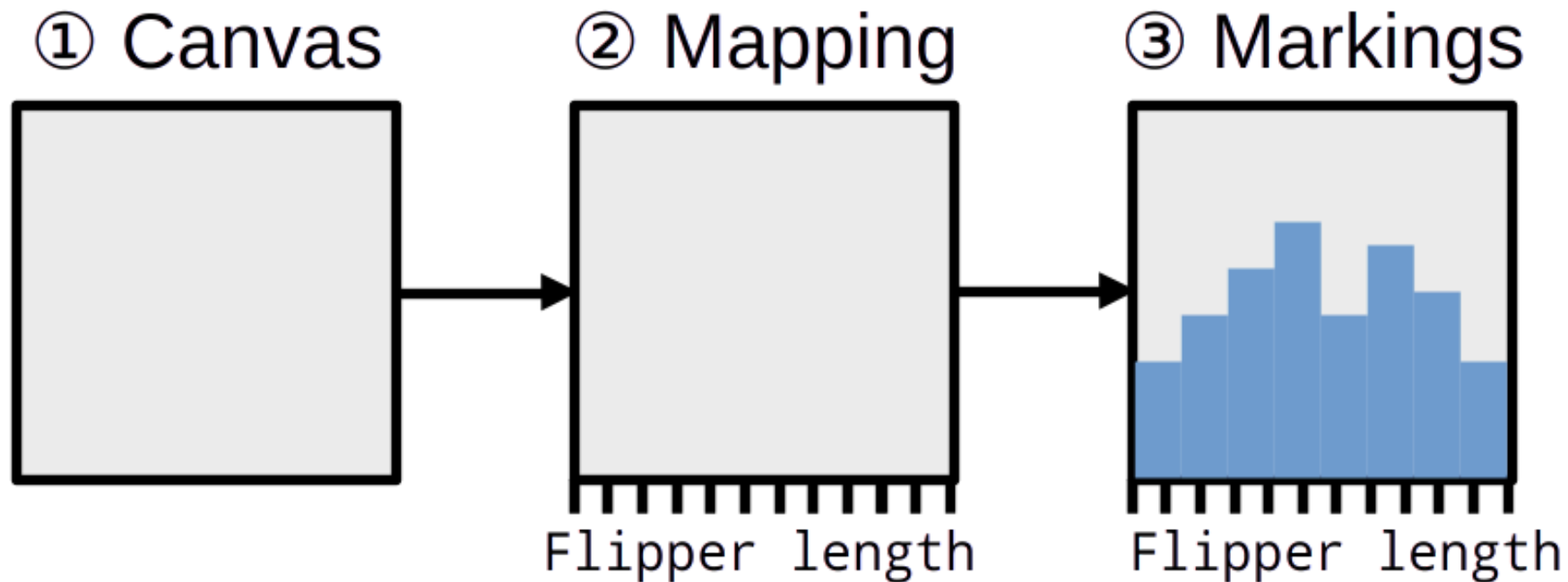
- Three species
- Measured at different locations/times
- Several physical characteristics + sex

How can we use this data to learn about the physical differences between the three species?

- Discuss ideas for a *visualization*
- Discuss ideas for a *numeric* summary
- What information is *lost* when we make these summaries?



The `tidyverse` package includes a “grammar of graphics” `ggplot()`.



## Single numeric variable

Histograms (`geom_histogram`), box plots (`geom_boxplot`)

## Single categorical variable

Bar graph (`geom_bar`), column graph (`geom_col`)

## Two categorical variables

Scatterplot (`geom_point`), line plot (`geom_line`)

`ggplot2` offers rich customization options for:

- Color
- Shape
- Size
- Transparency

and more...

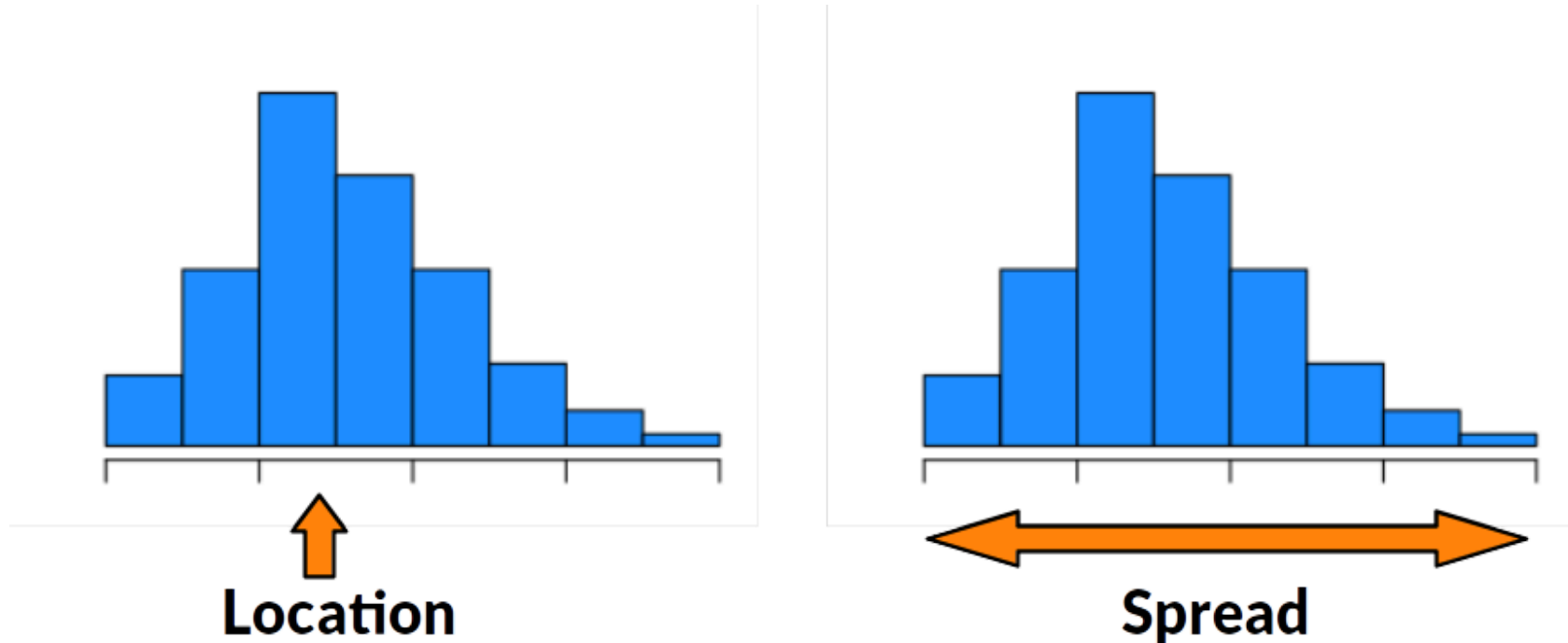
These don't change the *information* shown in the graph.

Choose one of the penguin data graphs we made and change the style from the default output.

- Change the color, size or transparency of at least one visual element.
- Look up the help file for the `geom_` function and figure out which aesthetic options are available.
- CHALLENGE: How is `color` different from `fill`?



Numeric summaries help quantify our data.



Location: mean, median, quantiles

Spread: sd, range, IQR

$\text{Mean} \equiv \bar{x} \equiv \frac{1}{n} \sum_{i=1}^n x_i$   
“*typical value*”

$\text{Standard deviation} \equiv s \equiv \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$  “*typical deviation from mean*”

The **median** is the midpoint of the data.

This generalizes to **percentiles**, e.g. 30th or 75th percentile.

Percentiles refer to the amount *below*!

SD is based on mean - other measures of spread rely on the percentiles.

The **range** is the “footprint” of the entire dataset.

The **inter-quartile range (IQR)** is the “footprint” of the middle 50% of the data.

Consider the following two statements:

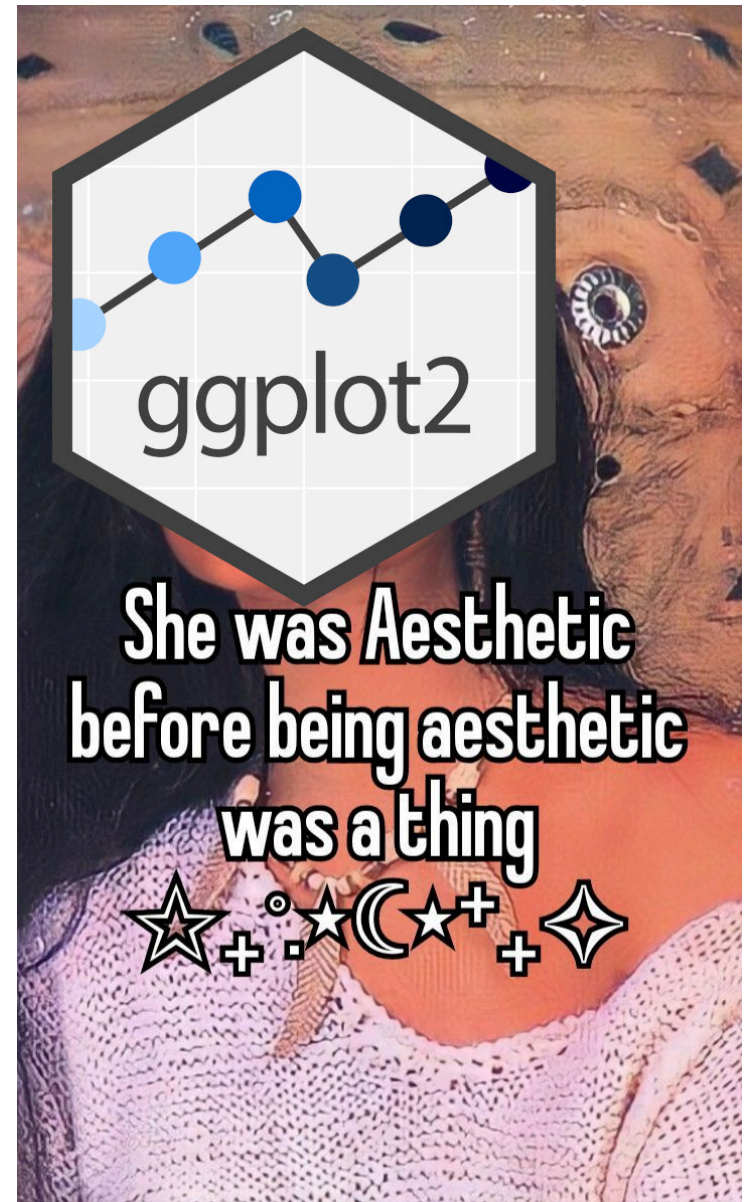
- “The average midterm score is 70/100”
- “Half of the students did worse than 70/100, and half did better”

Are these always both true? How could they be different?

Consider whether summaries are *robust* to outliers.

Let's look closer at ggplot  
aesthetics.

Color, fill, x, y, alpha, etc.



variable  
↓  
ggplot(aes(  )) +  
geom(  )  
↑  
constant

Aesthetics in our `aes()` mapping are **variable aesthetics**.

- Tied to a specific column (variable)

**Constant aesthetics** change the style of the plot.

- Not mapped to any column

What if we want a mapping to apply to only one layer?

Variable aesthetics can be either **global** or **local**.

(global) variable  
↓  
ggplot(aes(  )) +  
geom(aes(  ),  )  
↑  
(local) variable

The four `ggplot()` calls return either an error or an unexpected result. Identify and fix the problem with each one.

- Why does `geom_point()` not work?
- Why can't R find the `species` column?.
- Why are the points not huge? Why is there a legend for “size”?
- Why is there a blank gridded canvas?.



The `ggplot2` package offers lots of customization tools.

- Text annotations
- Axis scales + labels
- Color palettes

(Download the slide .pdf if the links don't work.)

## Looking ahead

In [chapter 5](#), we'll learn how to tidy and clean data. If you want a head start, run the following lines of code to create two different summary tables (make sure you have [tidyverse](#) and the penguin data loaded):

```
1 penguins %>%  
2   summarize(avg_flipper_len = mean(flipper_length_mm))  
3  
4 penguins %>%  
5   group_by(species) %>%  
6   summarize(avg_flipper_len = mean(flipper_length_mm))
```

What is the difference between the two summary tables? What do you think the `%>%` symbol is doing?